Tech Interview Prep:

**Approximate Nearest Neighbor (ANN) Mechanics & Strategic Chunking Architecture**

“How does an approximate nearest neighbor (ANN) search work, and how do you choose optimal chunk sizes for technical versus narrative documentation?”

1) The Mathematical & Algorithmic Mechanics of ANN (HNSW & IVF-PQ)
2) The Recall-Latency-Memory Trade-off Frontier
3) Strategic Chunking Architectures: Technical APIs vs. Narrative Prose
4) Advanced Production Patterns (Parent-Child & Contextual Ingestion)

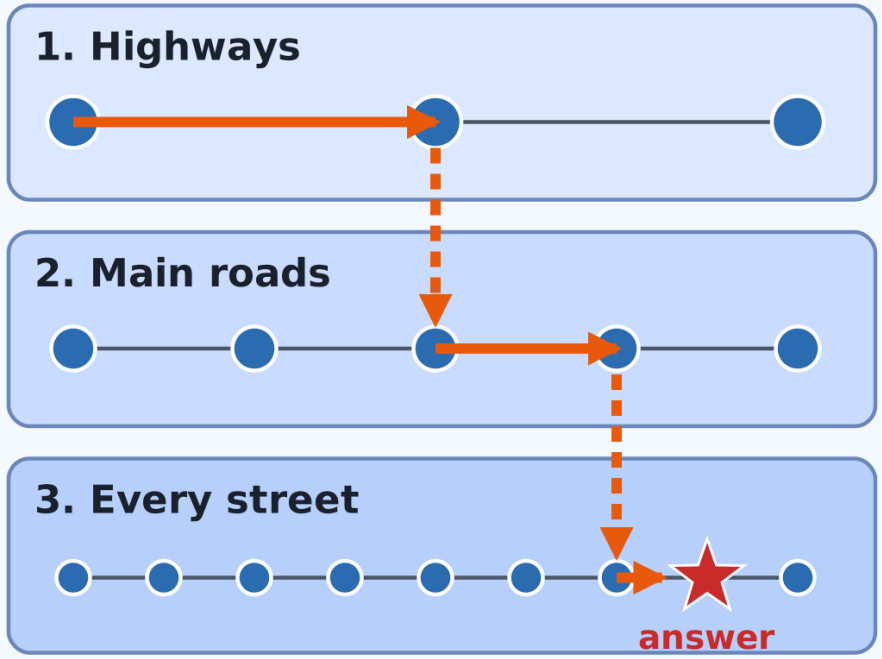
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# ANN Search Mechanics (HNSW vs. IVF-PQ)

## Two Ways to Search Faster (ANN)

### HNSW = the map method

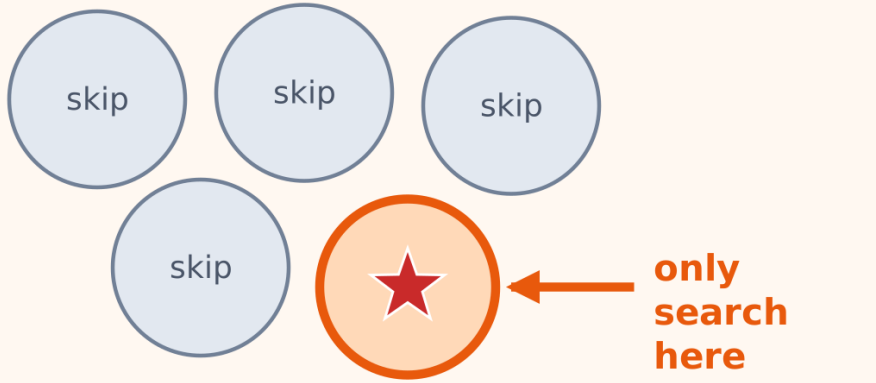



**Zoom in step by step**

Fast, but uses a lot of memory

### IVF-PQ = the shortcut method

**Step 1: only check the closest group**



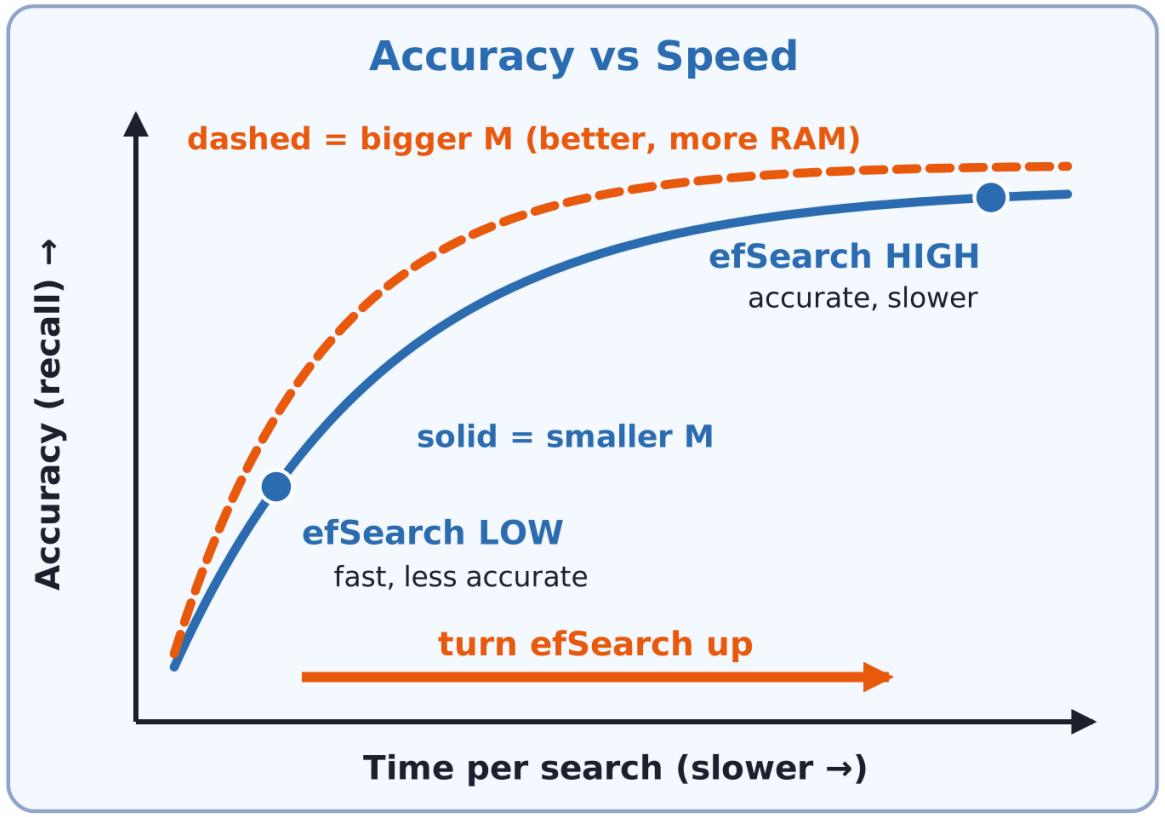

**Step 2: shrink the data (PQ)**

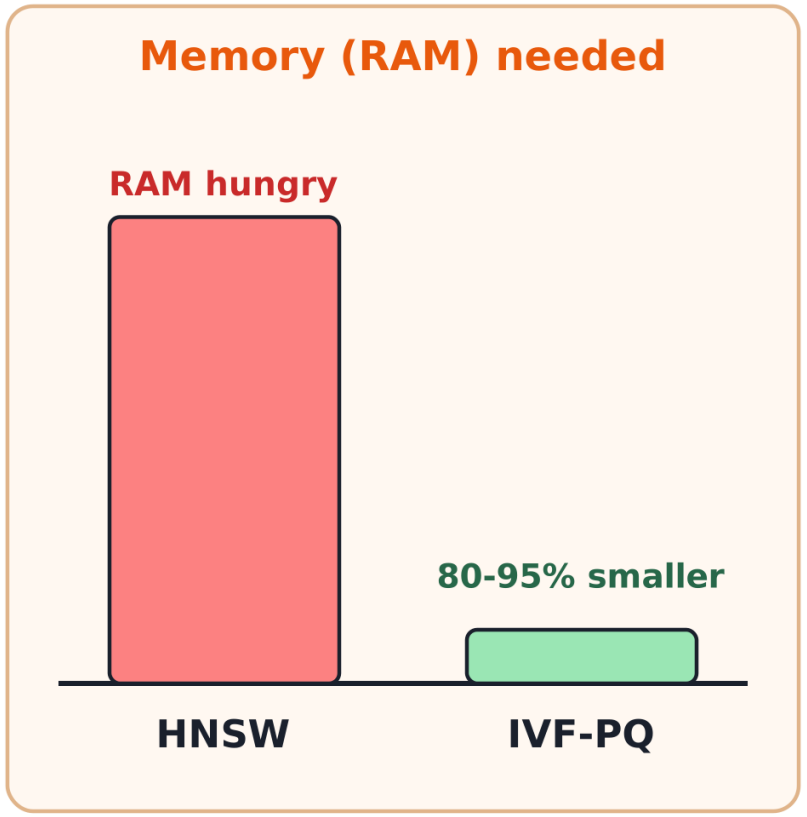



Much less memory, slightly less exact

# The Production Trade-Off Frontier: Memory, Recall & Tuning

## The Trade-Off: Speed vs Accuracy vs Memory




**M = links per dot**

More links = more accurate,
but more RAM, slower build

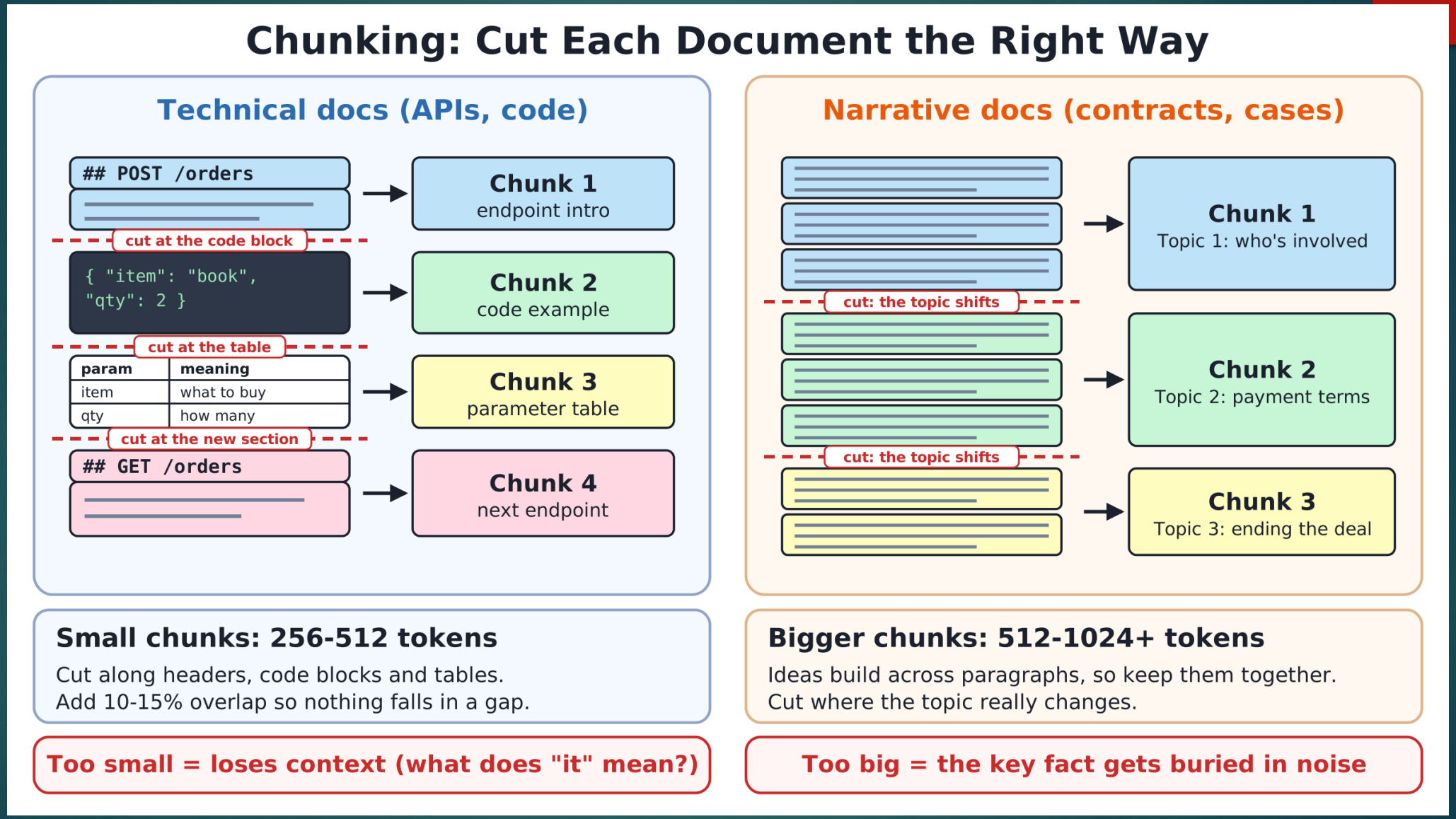
**efConstruction = build care**

Higher = better graph,
but slower to build

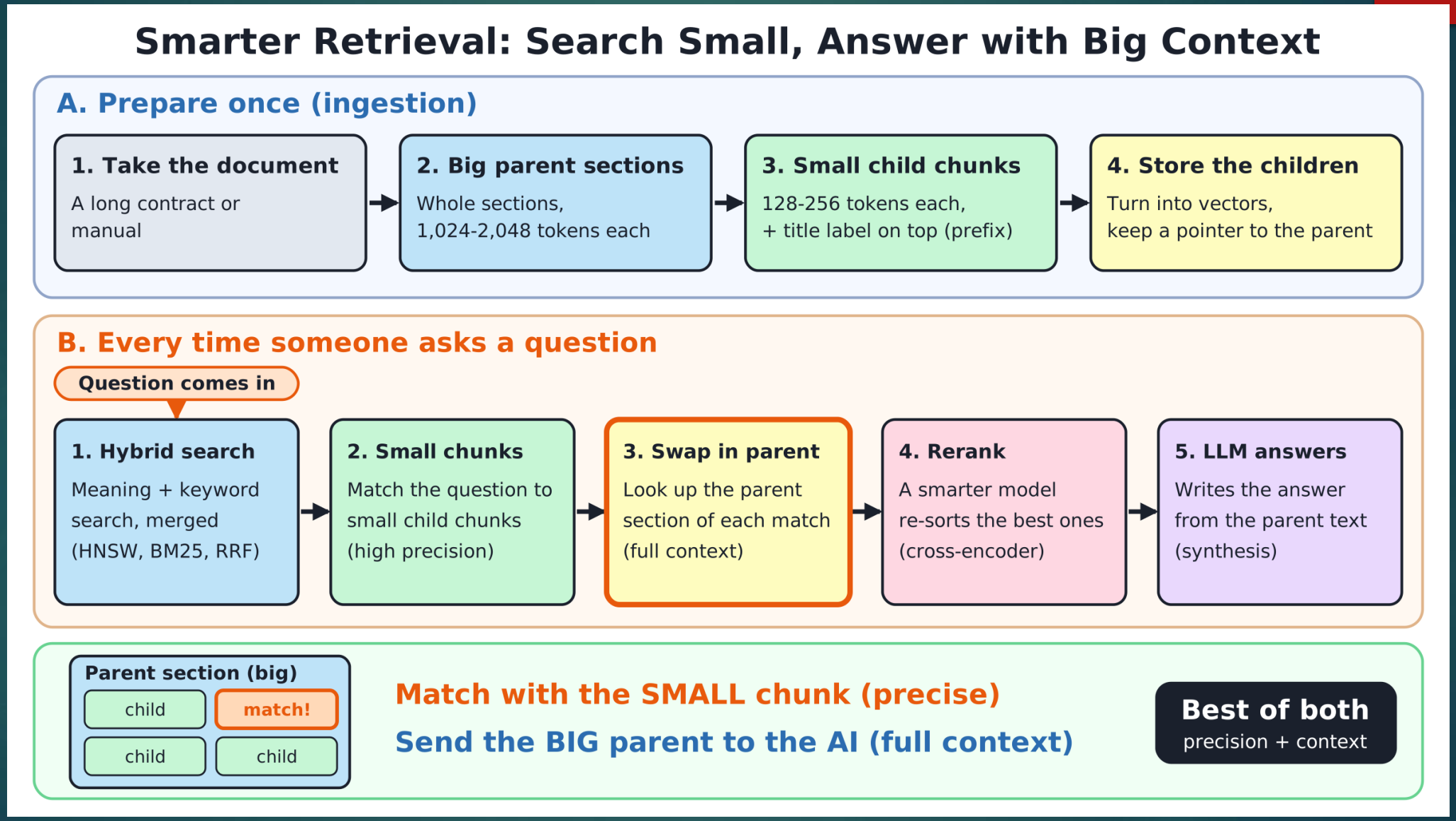
**efSearch = search effort**

Higher = more accurate,
but slower (live slider)

# Strategic Chunking: Technical vs. Narrative Documentation



# Advanced Mitigations: Decoupling Retrieval from Context



# Summary & Systems Architecture Takeaways

- Understand the trade-off: speed vs. accuracy vs. memory
- Cut documents based on types of documents
- Search with small pieces but answer with big ones